5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

code;

```
import cv2
import matplotlib.pyplot as plt

def analyze_histogram(path):
    # Read the image
    img = cv2.imread(path)

    if img is None:
        print("Image not found!")
        return

    # Define colors and channel names
    colors = ('b', 'g', 'r')
    names = ('Blue', 'Green', 'Red')

    # Calculate and plot histograms
    for i, color in enumerate(colors):
        hist = cv2.calcHist([img], [i], None, [256], [0, 256])
        plt.plot(hist, color=color, label=names[i])

    plt.title("Color Histogram")
    plt.xlabel("Pixel Intensity (0-255)")
    plt.ylabel("Number of Pixels")
    plt.legend()
    plt.show()
```

```
# Call the function
analyze_histogram(
    r"C:\Users\ROHIT\Pictures\Instagram\IMG_20230103_200212_526.jpg"
)
```

output

